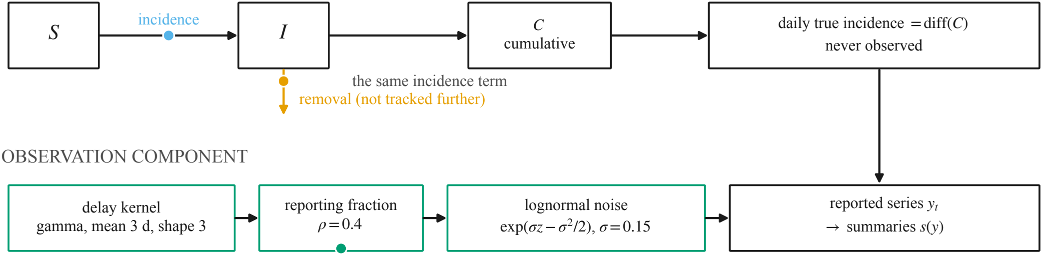
DETERMINISTIC CORE — fixed-step RK4, 24 substeps/day, $T = 120$ days, $N = 10^5$



a CONTINUOUS multiplicative layer is required: a count-valued one is not differentiable under common random numbers, and is kept as a negative control

S_A: peak height, peak time, final size, growth rate ($d = 4$)

S_B: 10 equal-width time-binned incidence counts ($d = 10$)

S_C: final size, peak height ($d = 2$, impoverished control)

THE THREE DISTORTION FAMILIES — one one-parameter family per component, in two declared sets

η_1 TRANSMISSION	base:	$\beta SI/N \div [1 + \eta_1 (I/N)/p_{\text{ref}}]$	saturating incidence — a prevalence nonlinearity
	adv:	$\beta \to \beta e^{\eta_1}$	constant multiplier, aimed at progression through R_0
η_2 PROGRESSION	base:	$\gamma \to \gamma e^{\eta_2 (t/T - 1/2)}$	mean-centred hazard drift — a timing distortion
	adv:	$\gamma \to \gamma e^{-\eta_2}$	sign-aligned with η_1', so both raise R_0
η_3 OBSERVATION	base:	$\rho \to \rho e^{\eta_3}$	pure amplitude error
	adv:	$\rho \to \rho e^{\eta_3 (t/T - 1/2)}$	reporting trend; adds η_3/T to the fitted growth rate

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every family is the base simulator EXACTLY at $\eta_k = 0$, asserted bit-for-bit; η_k is in units of ETA_SCALE $= 0.1$, a 10\% relative deformation

under the observation family the delay kernel and σ are HELD FIXED and only ρ moves — a stated limitation of the design, not an omission here